

Substructures Induced by Dust Drag

SI developed in the viscous gas

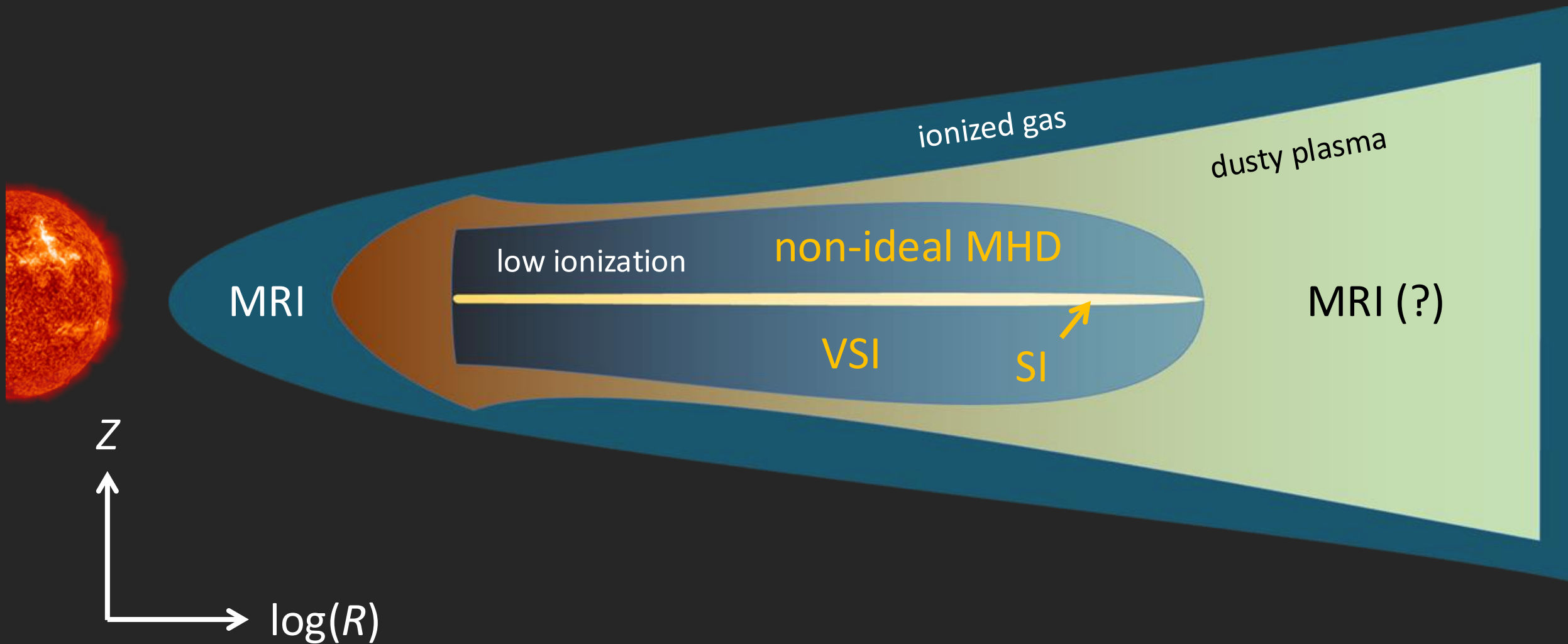
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Bi et al. 2026, A&A, 710, A21

SI is not a spatially isolated process

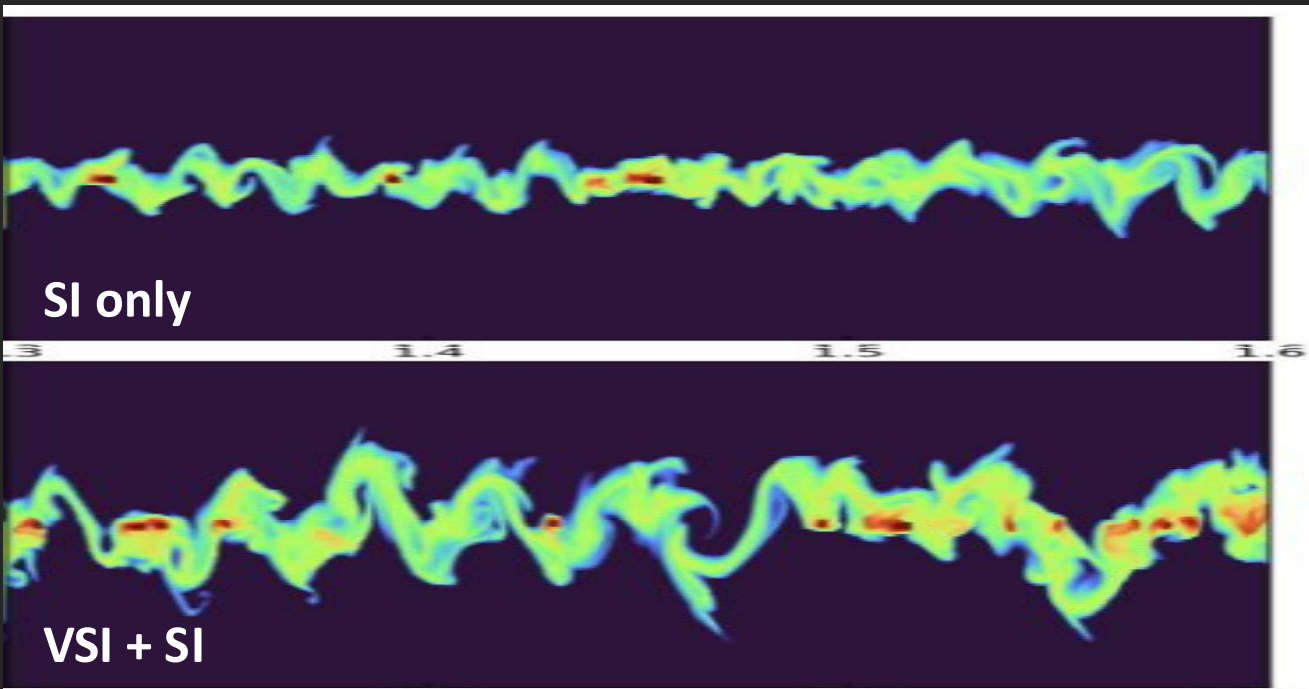


SI under VSI-driven turbulence

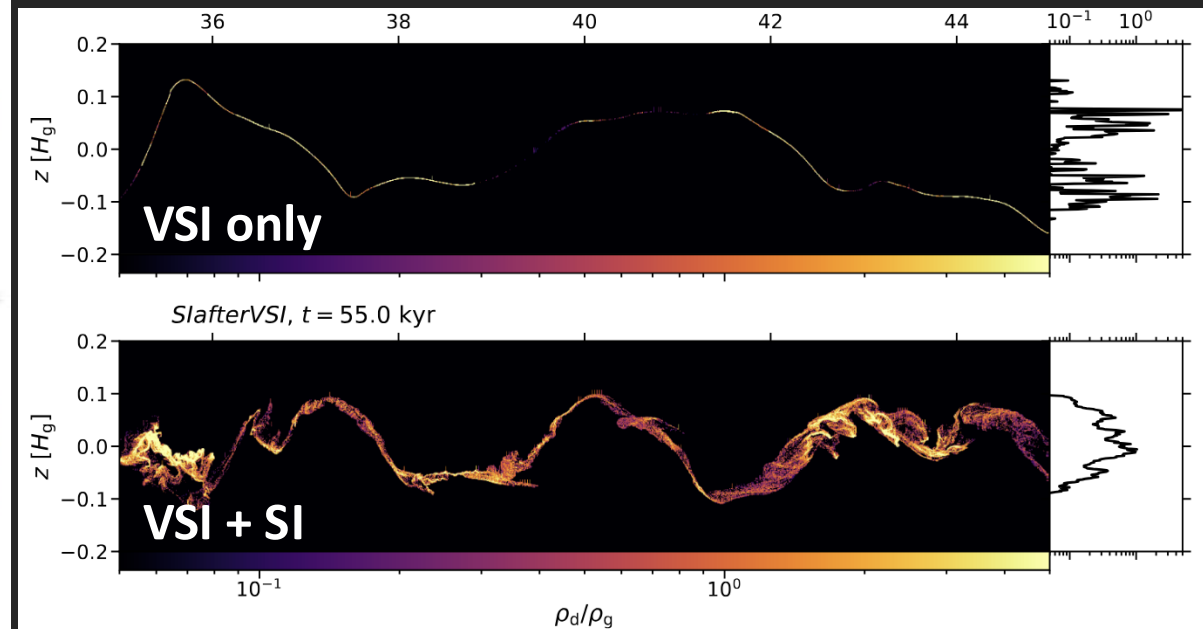
VSI-driven turbulence has an impact on the dust dynamics and SI

Huang & Bai, 2025a, 2025b

Schäfer et al., 2020, 2022, 2025



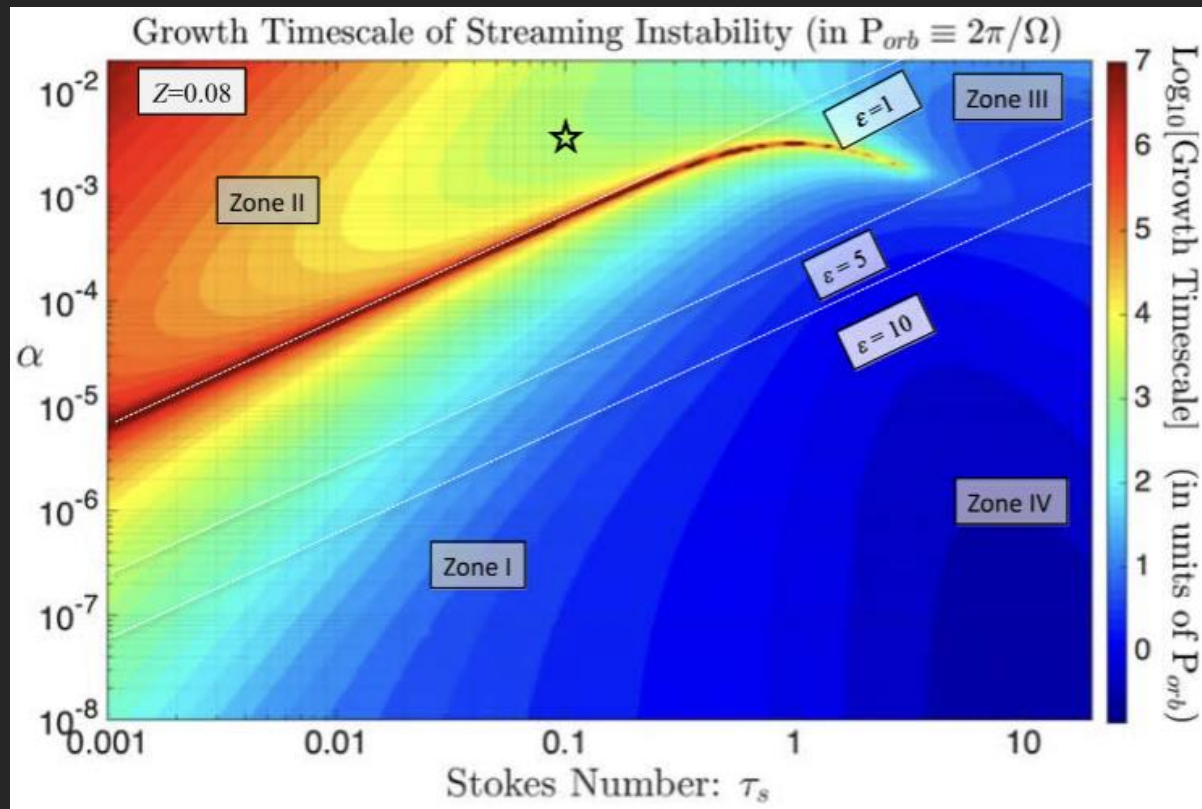
Huang & Bai, 2025a



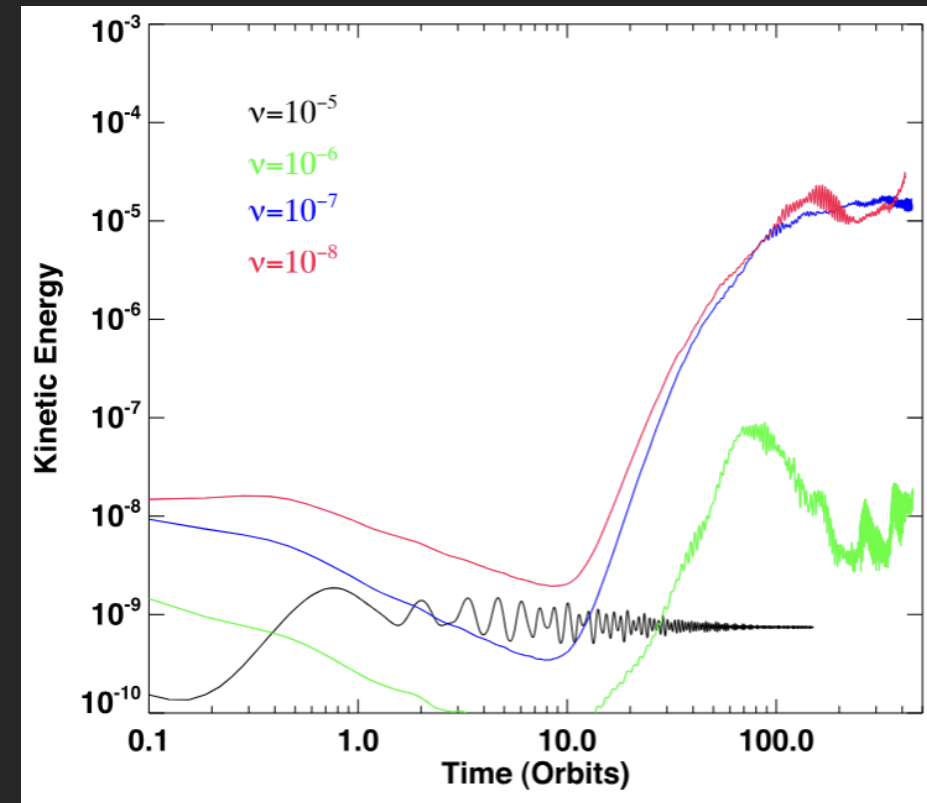
Schäfer, Johansen, & Flock, 2025

Viscosity disfavors SI and VSI

$$\alpha \lesssim \frac{Z^2}{1 - Z^2} \text{St} \simeq Z^2 \text{St} \quad (\text{Chen \& Lin, 2020})$$



Umurhan, Estrada, & Cuzzi, 2020

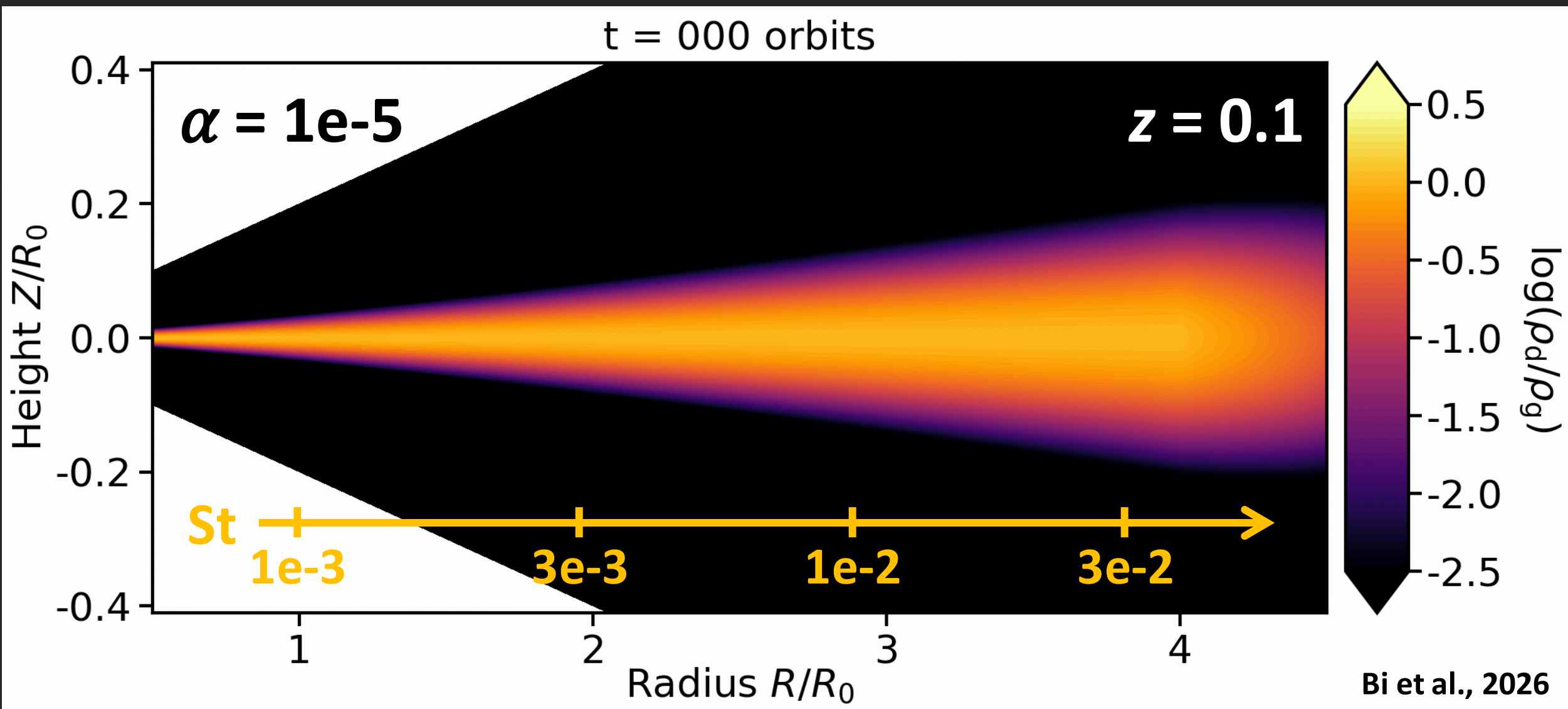


Nelson, Gressel, & Umrhan, 2013

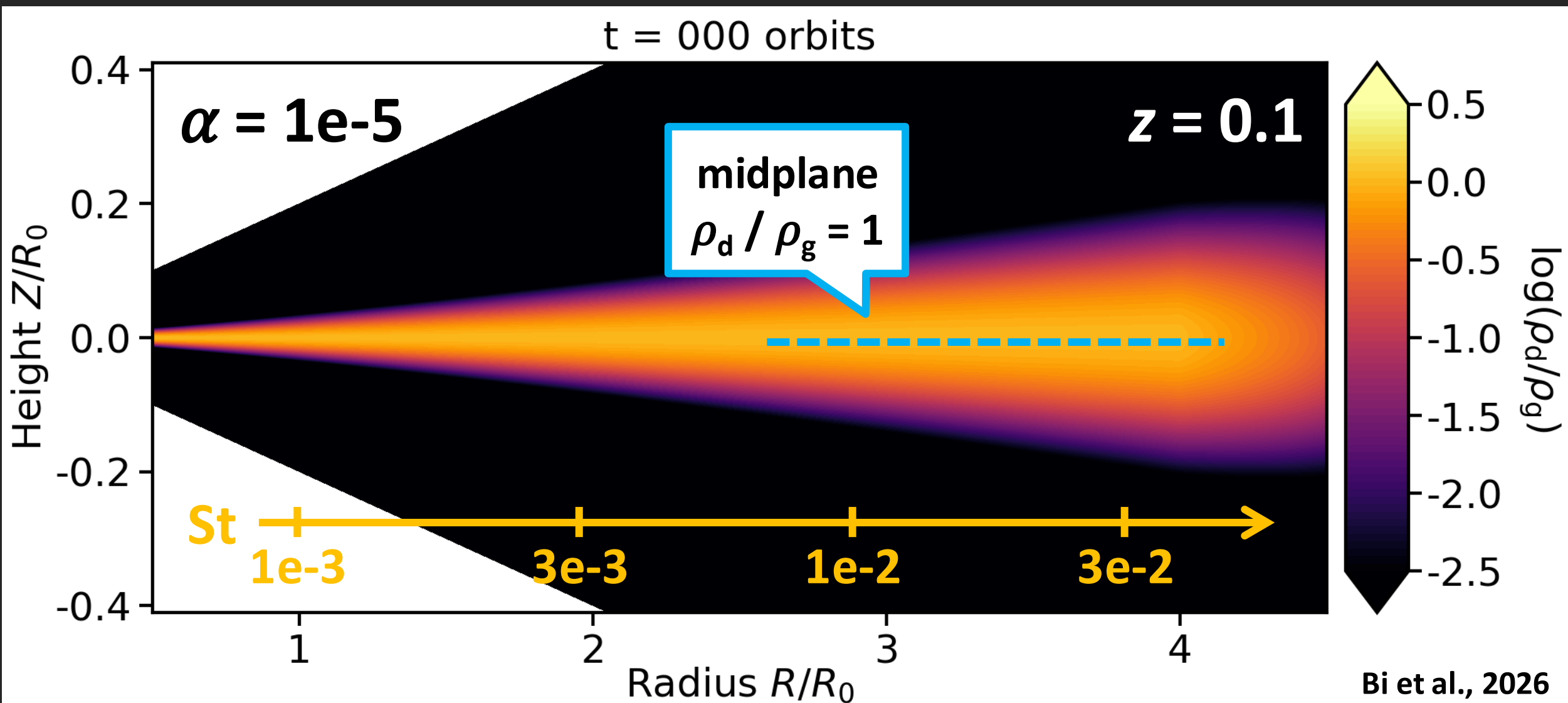
$$\begin{aligned} \log Z_{\text{crit}} = & 0.15 \times \log \alpha \times \log \alpha \\ & - 0.24 \times \log \alpha \times \log \tau_s \\ & + 1.18 \times \log \alpha \\ & - 1.48 \times \log \tau_s \end{aligned}$$

Lim et al., 2024

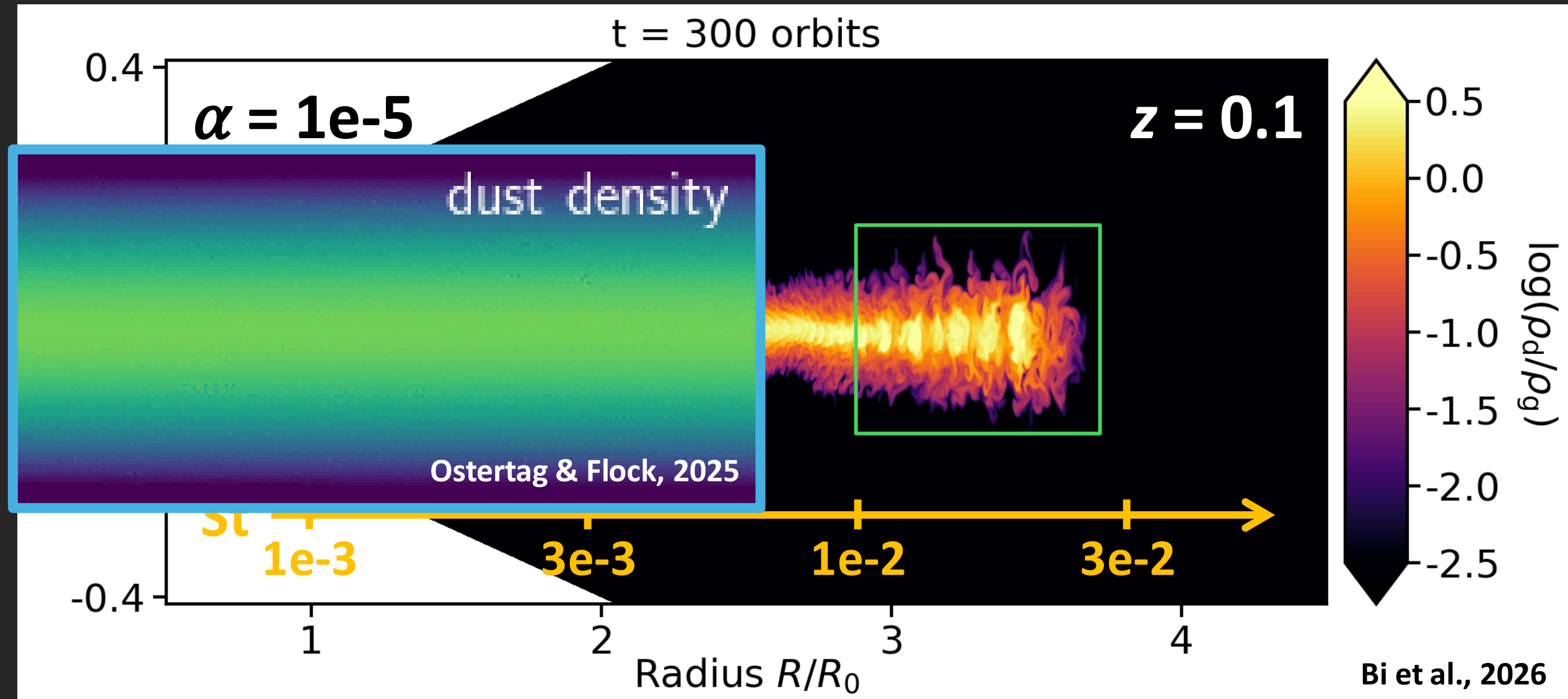
SI + VSI in the low-viscosity gas ($\alpha = 1e-5$)



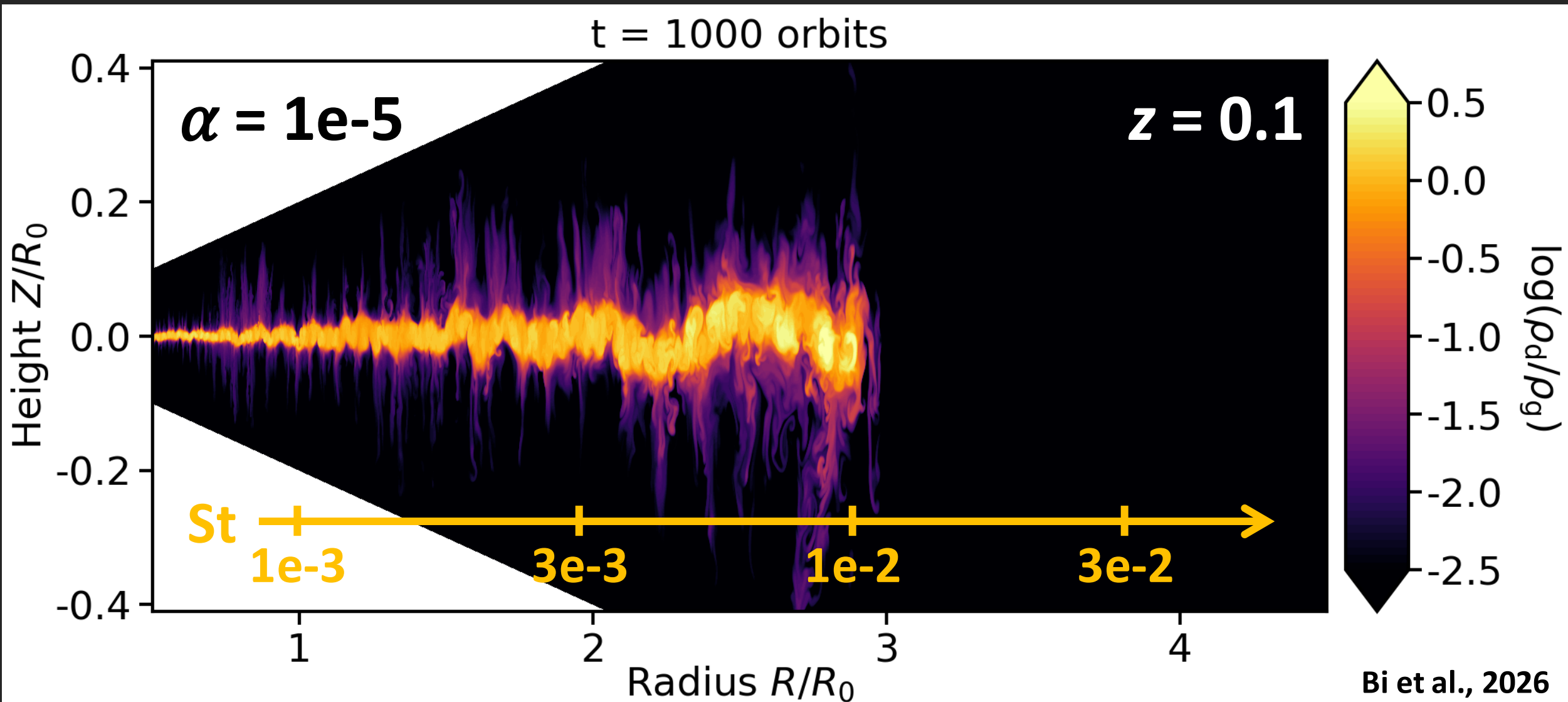
SI + VSI in the low-viscosity gas ($\alpha = 1e-5$)



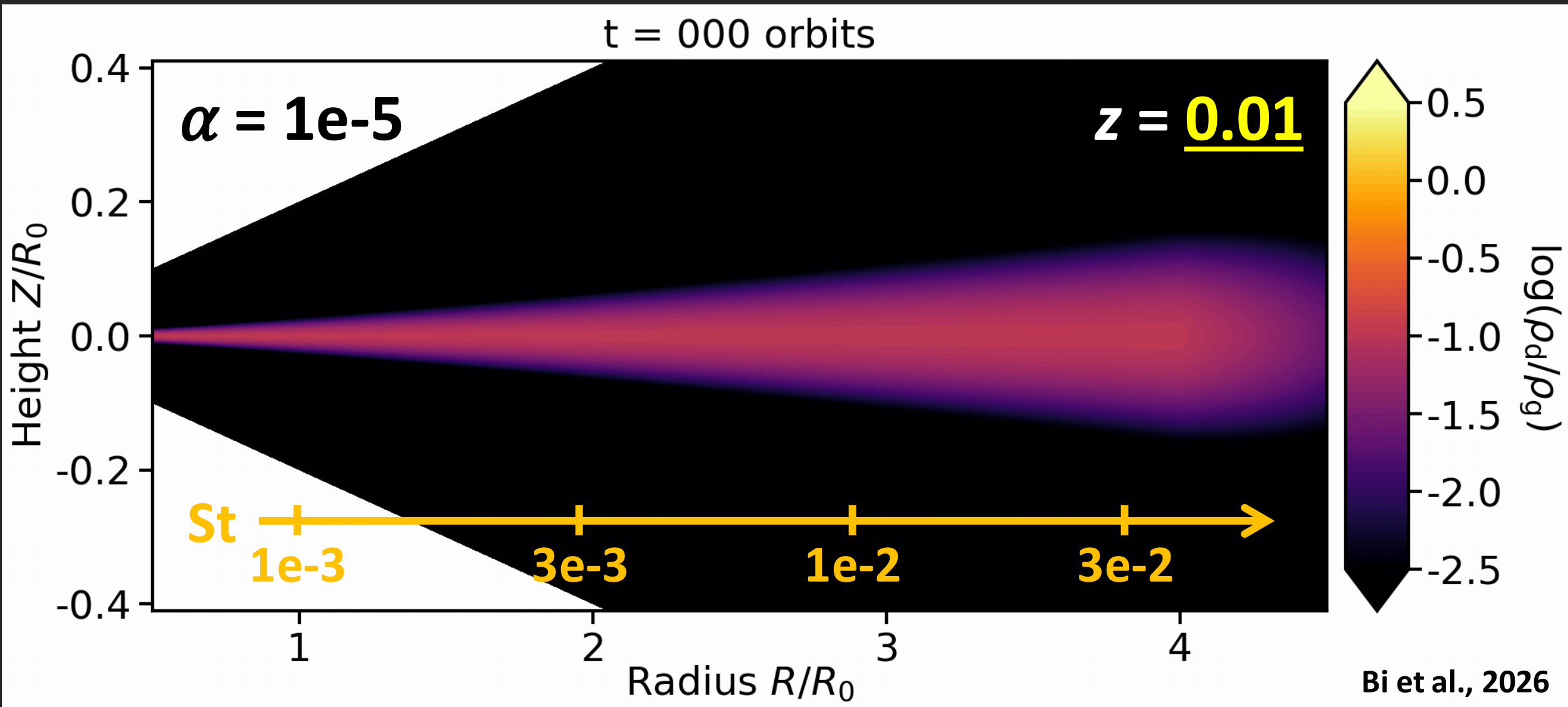
SI + VSI in the low-viscosity gas ($\alpha = 1e-5$)



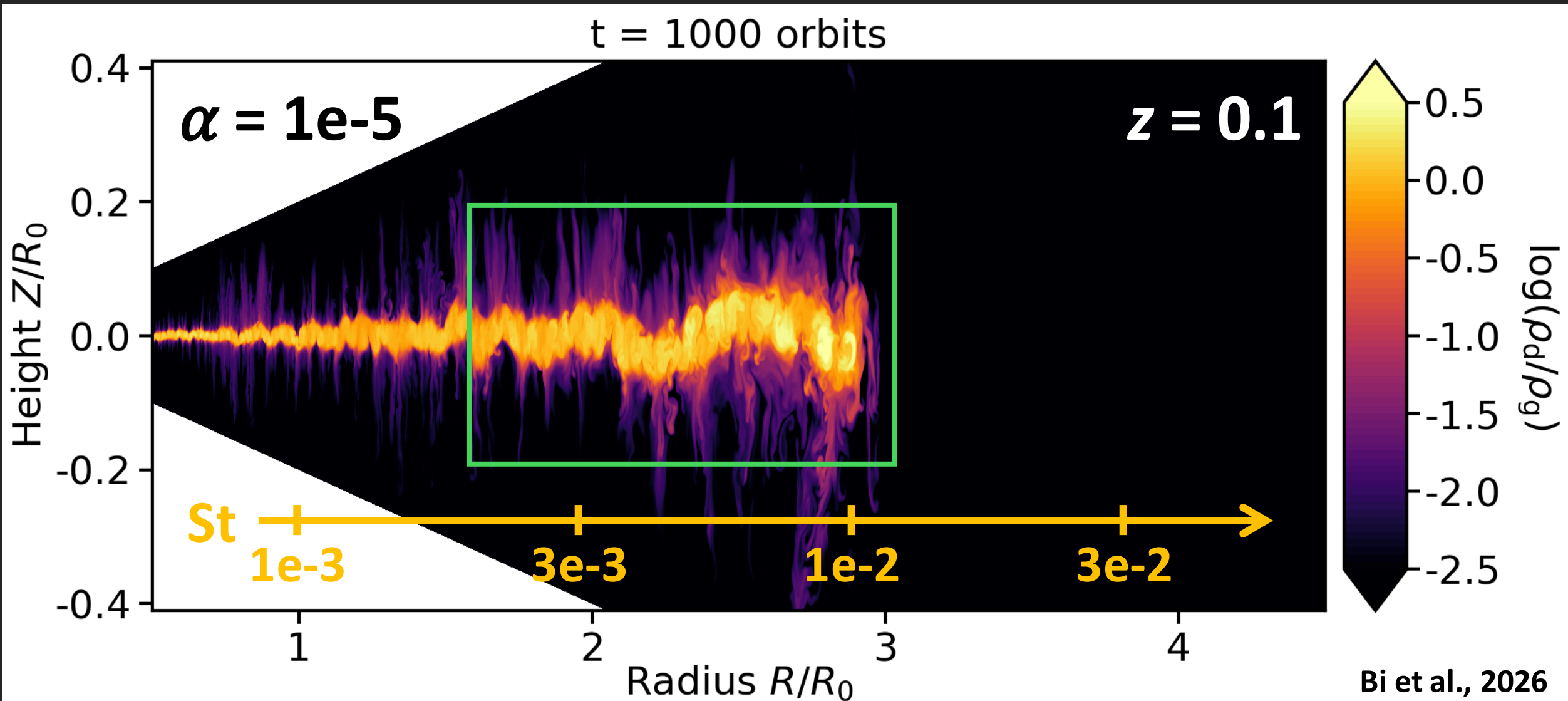
SI + VSI in the low-viscosity gas ($\alpha = 1e-5$)

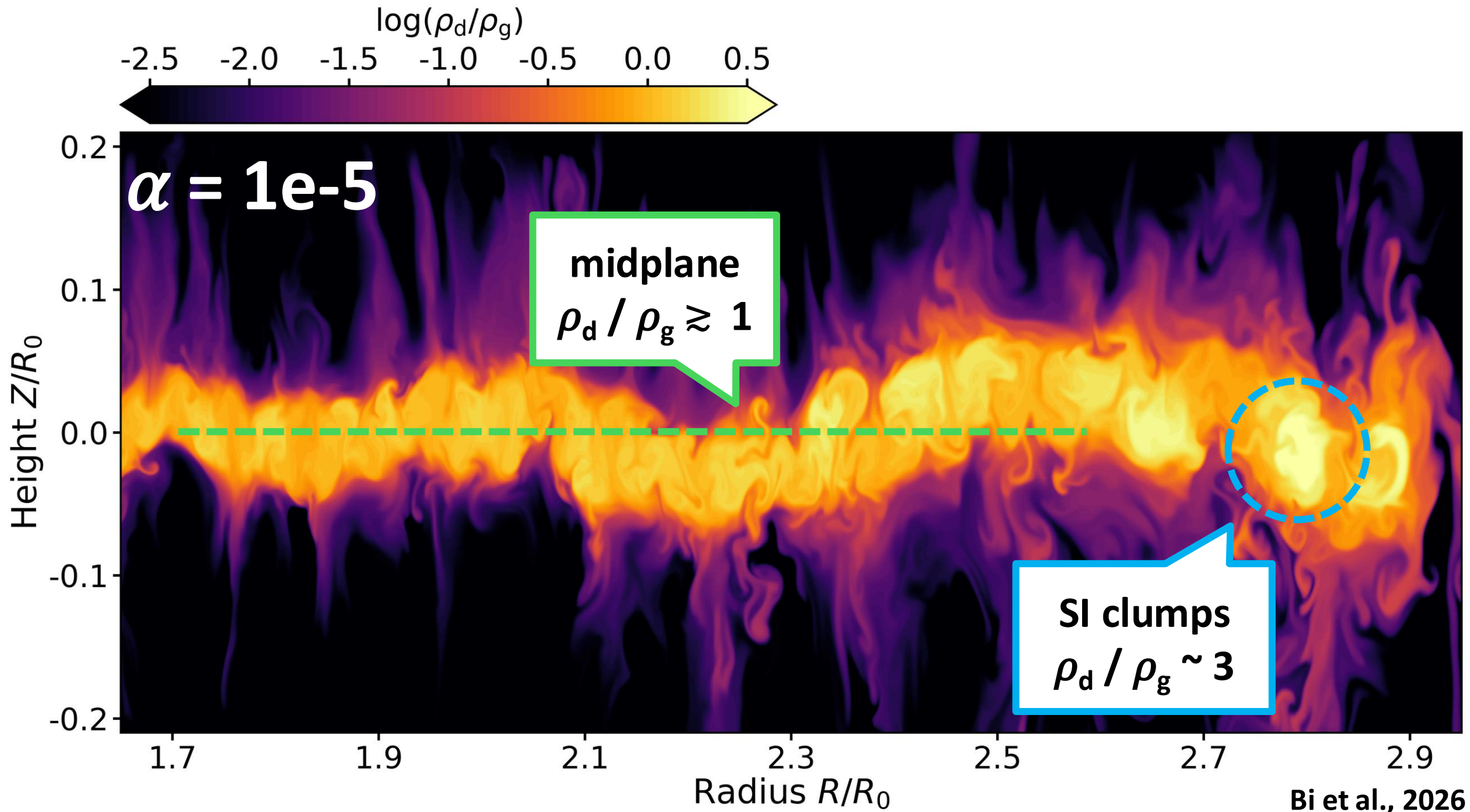


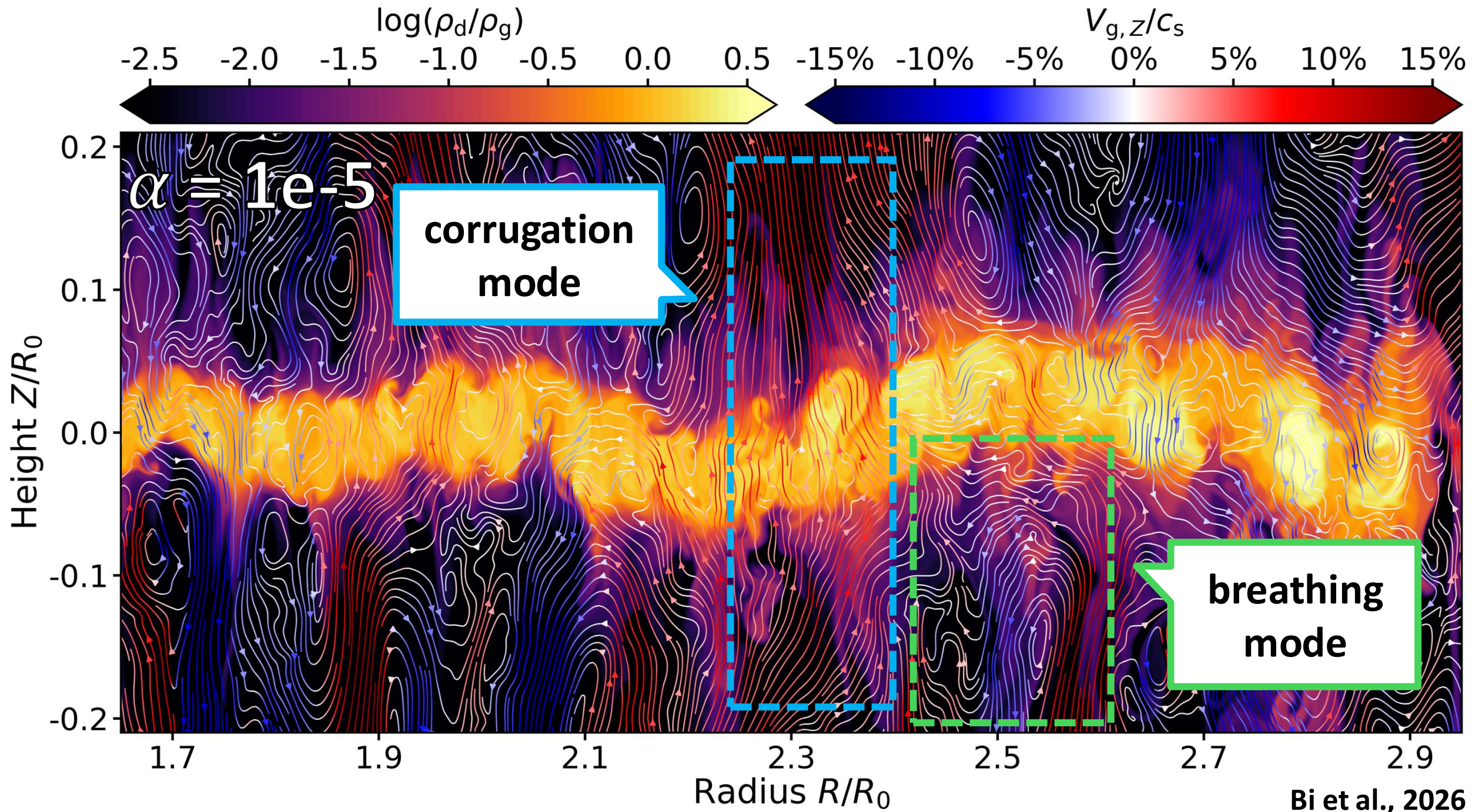
SI overwritten by VSI w/ low metallicity



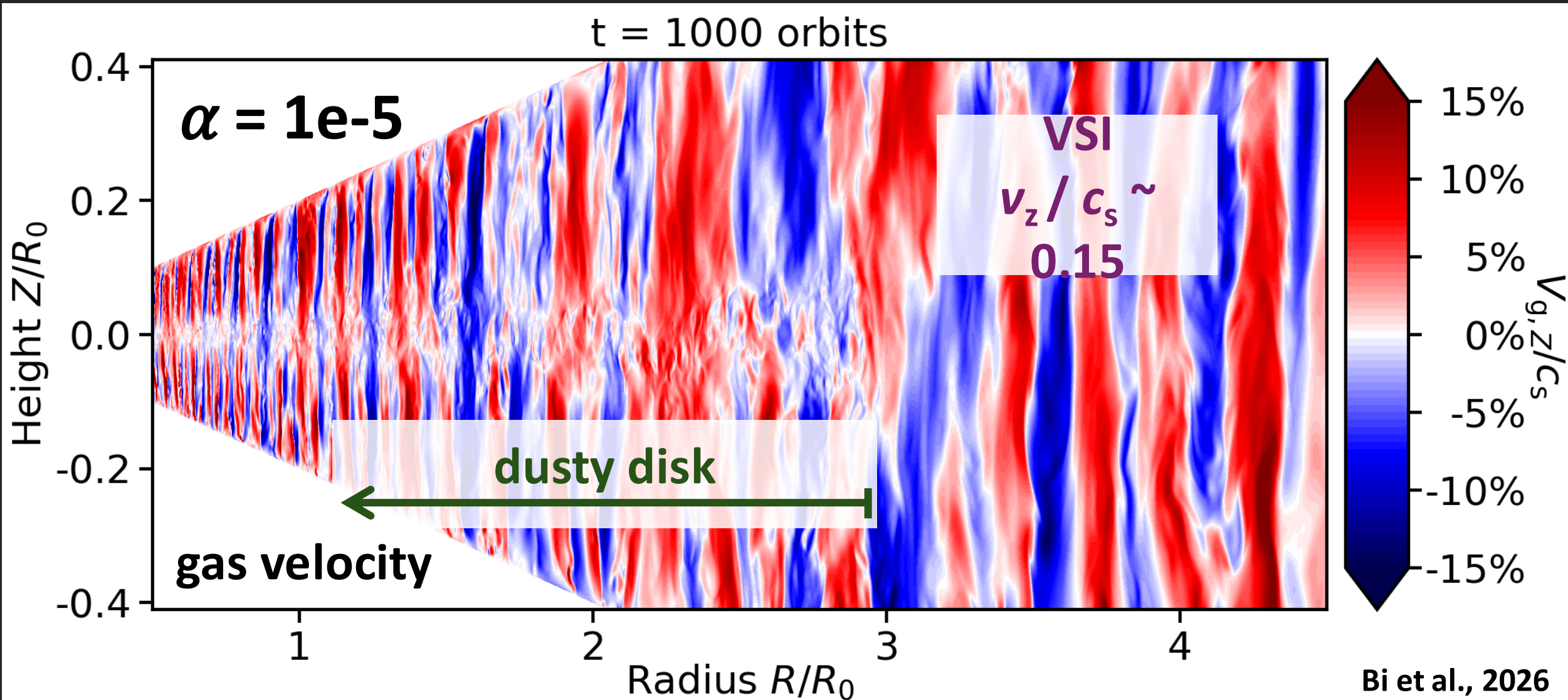
SI + VSI in the low-viscosity gas ($\alpha = 1e-5$)



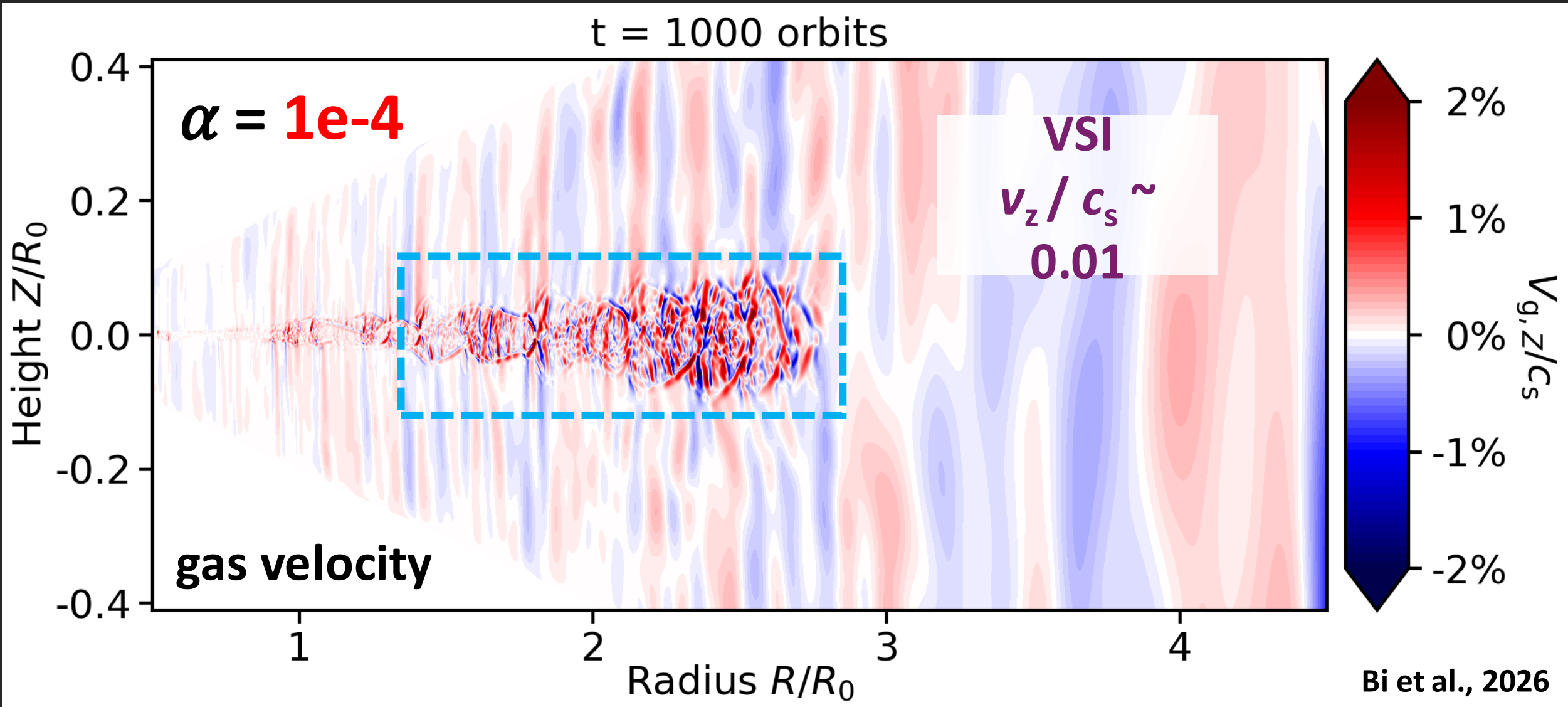




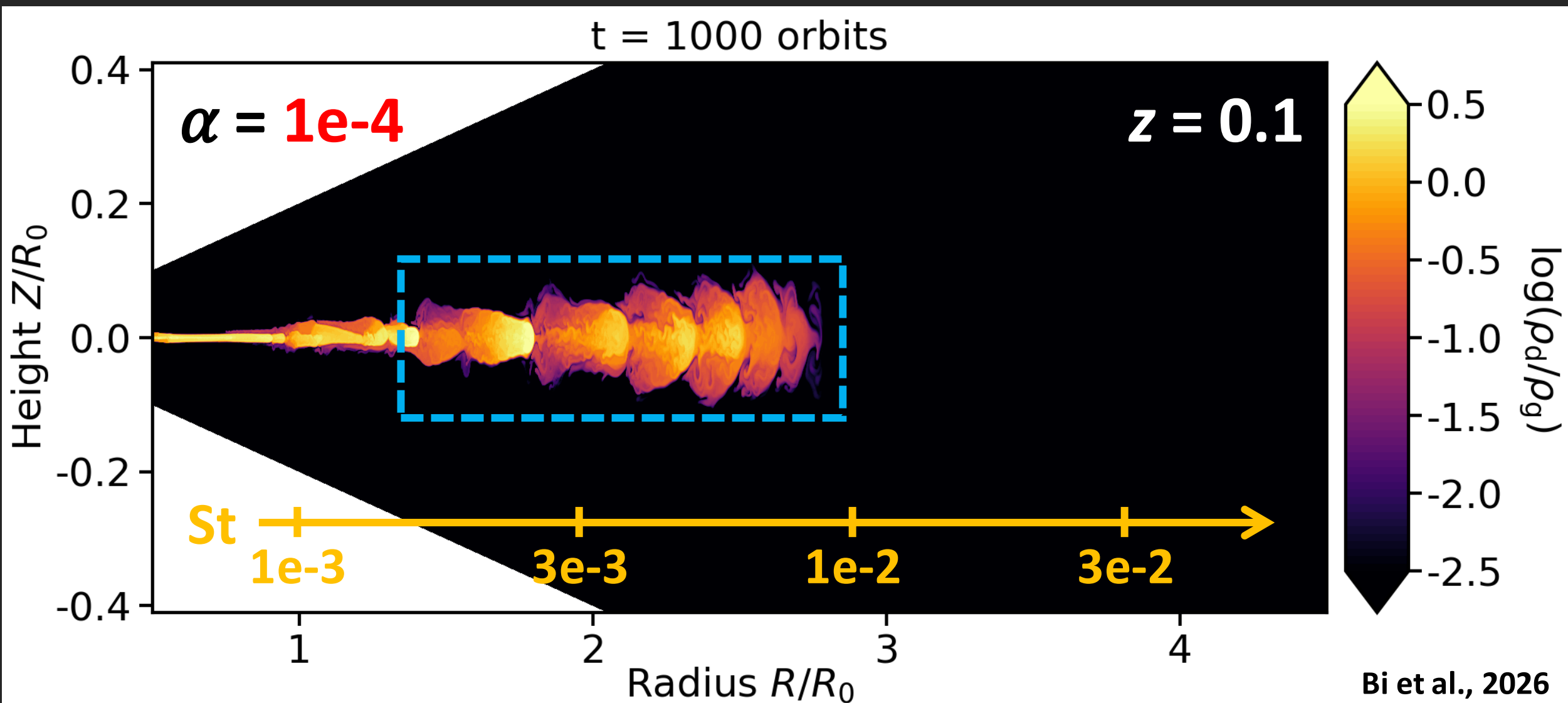
Gas velocity in the low-viscosity gas ($\alpha = 1e-5$)



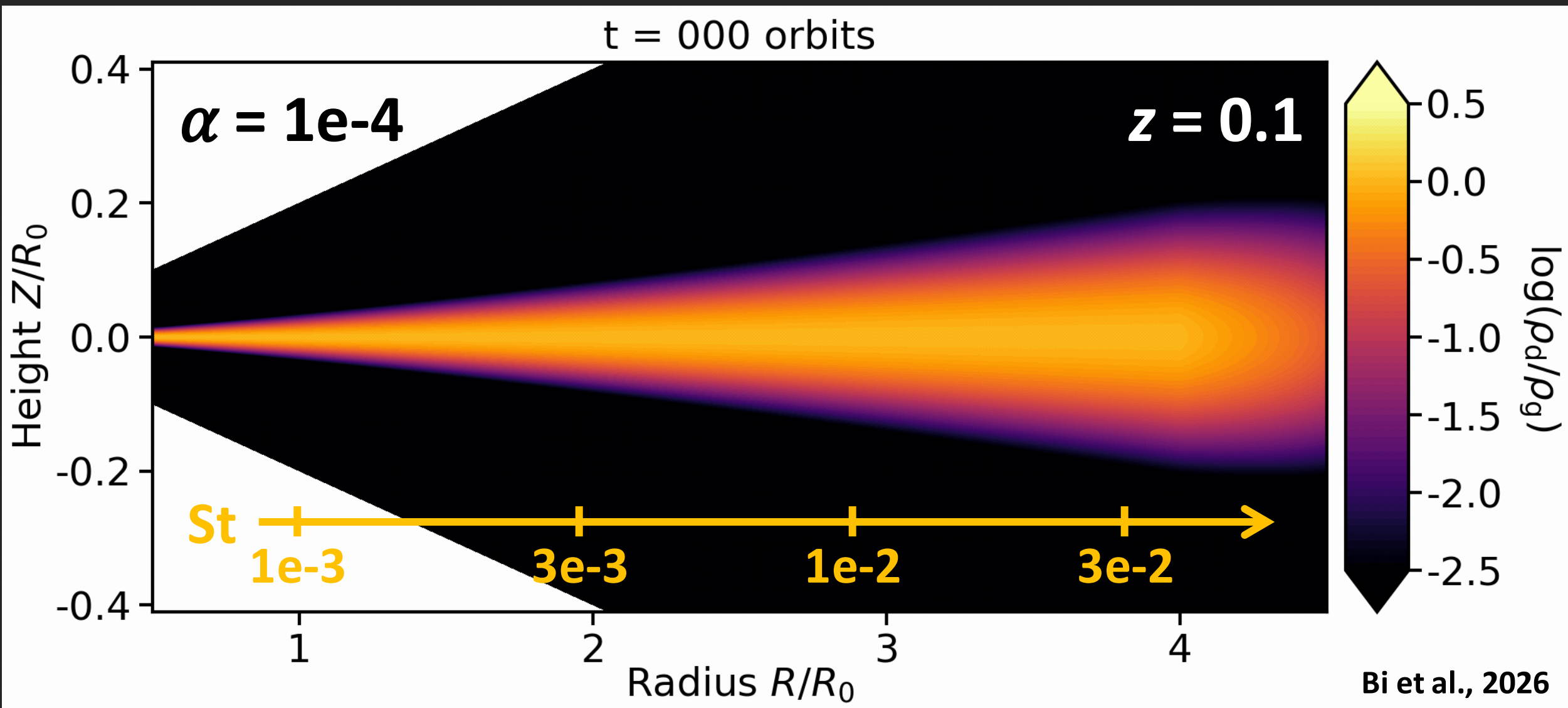
Gas velocity in more viscous gas ($\alpha = 1e-4$)

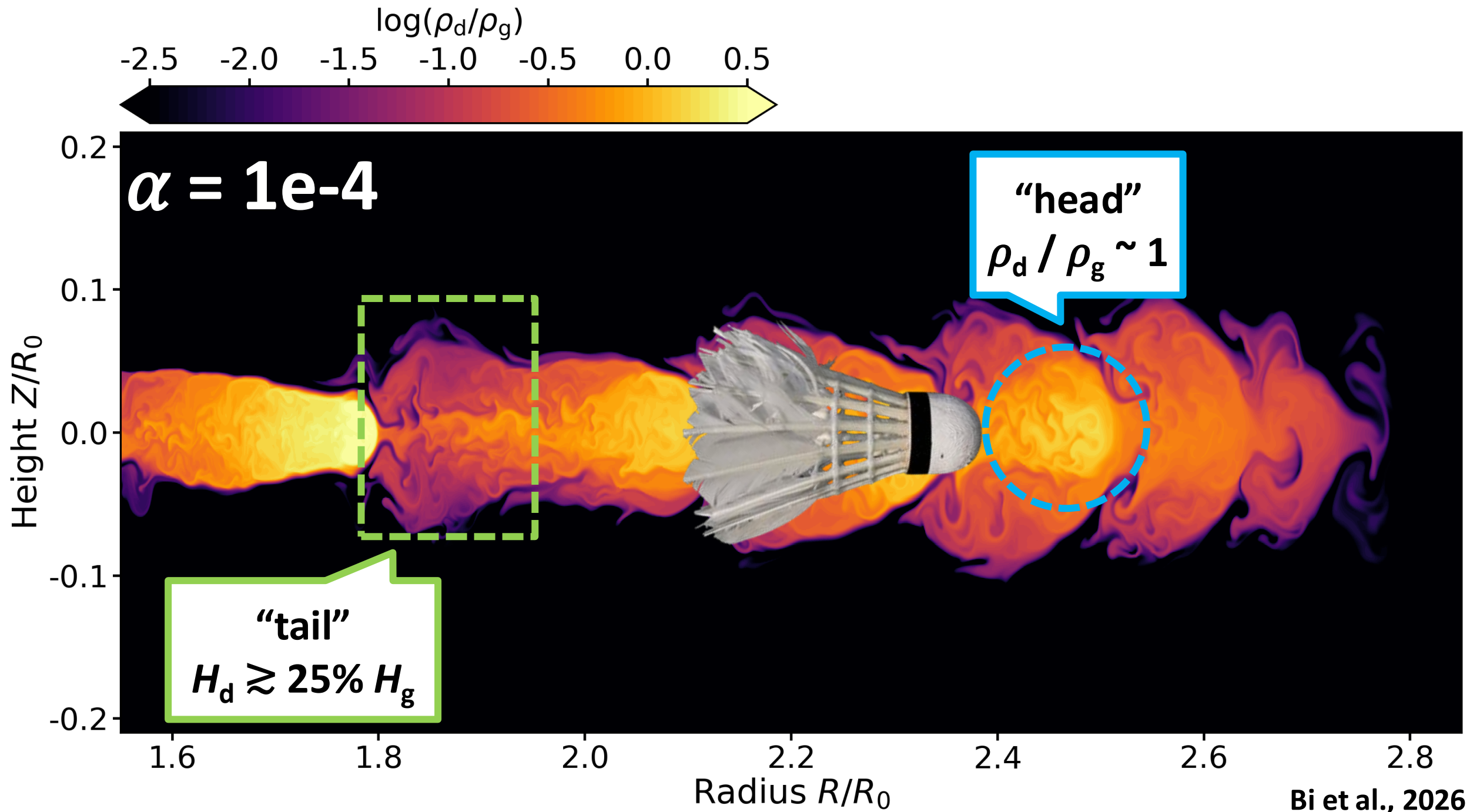


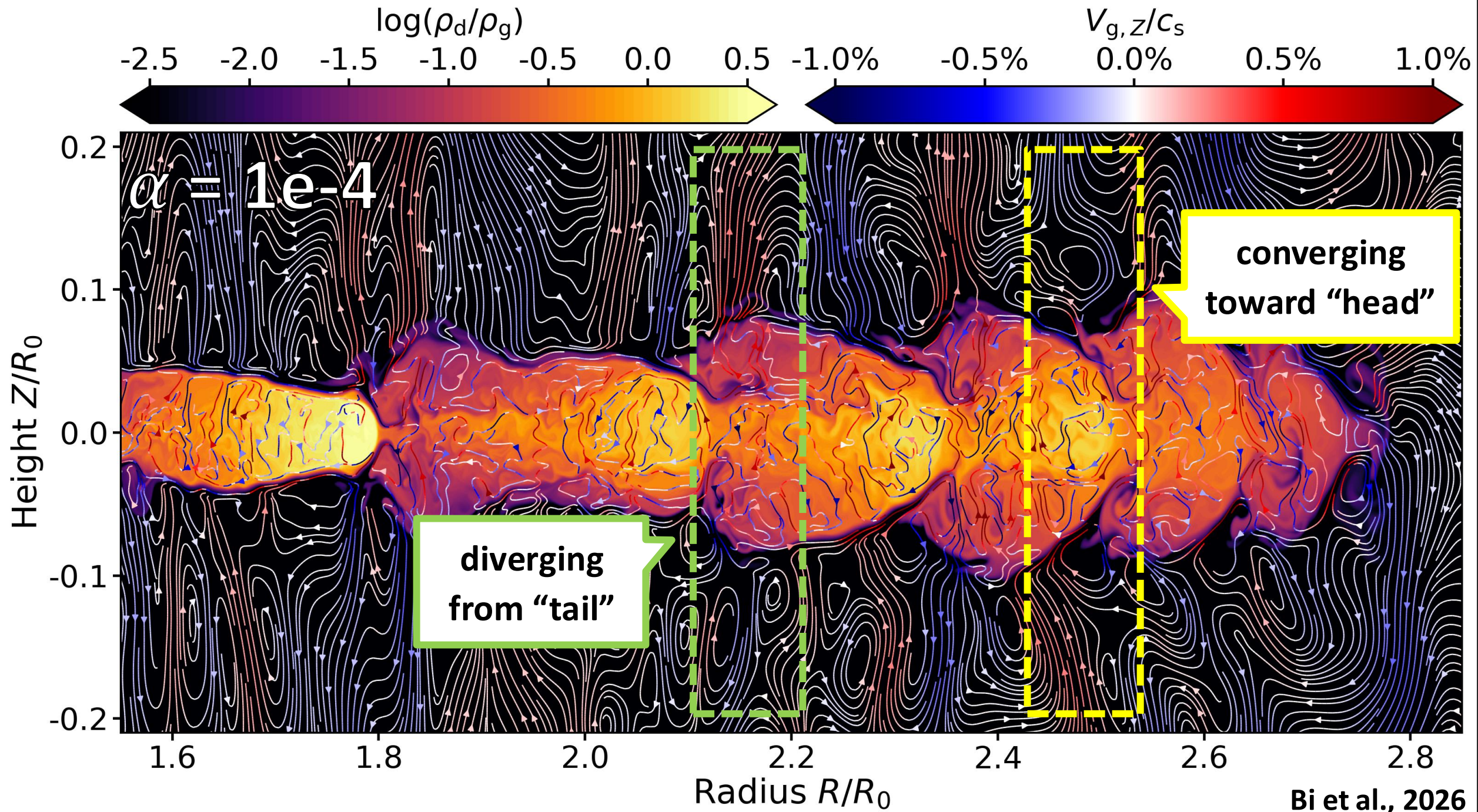
Dust substructures in more viscous gas ($\alpha = 1e-4$)



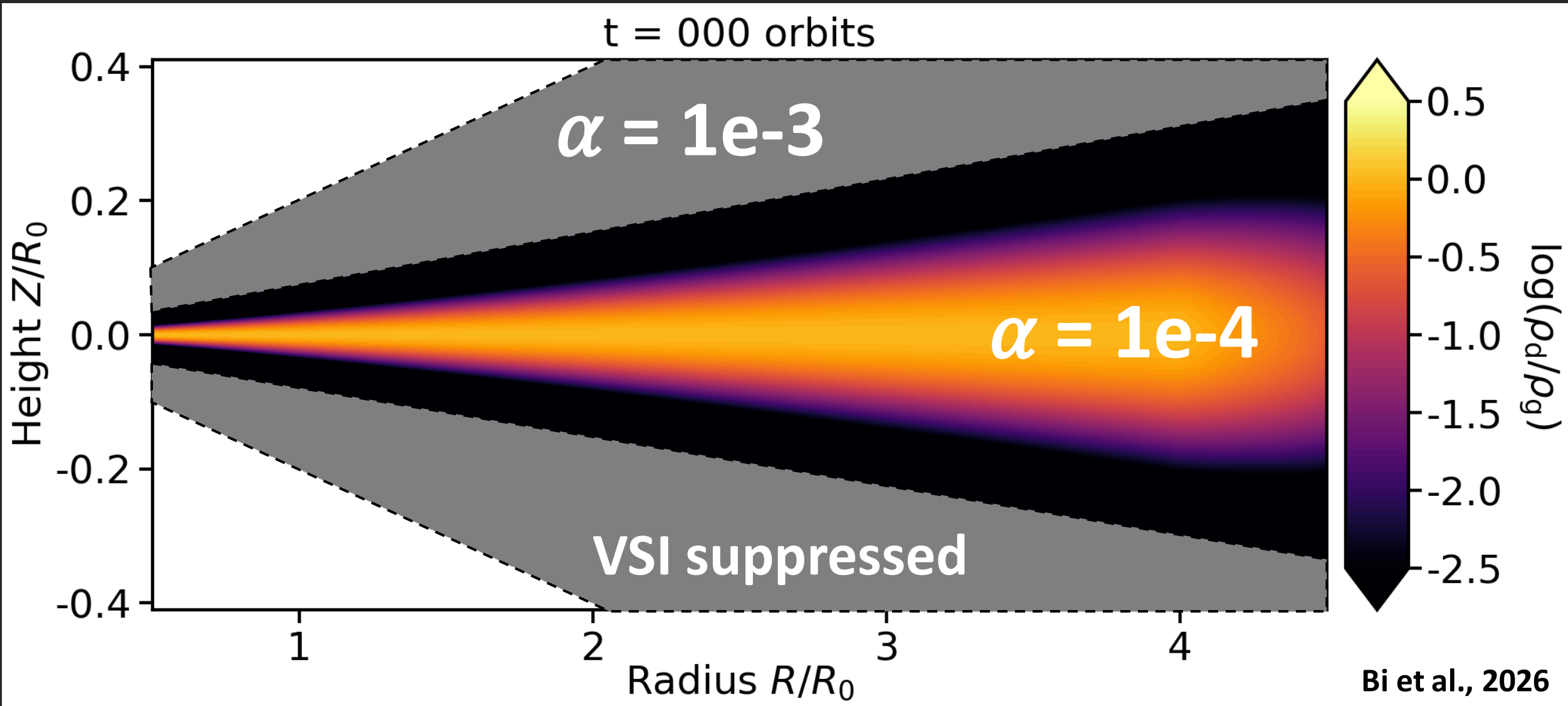
Dust substructures in more viscous gas ($\alpha = 1e-4$)



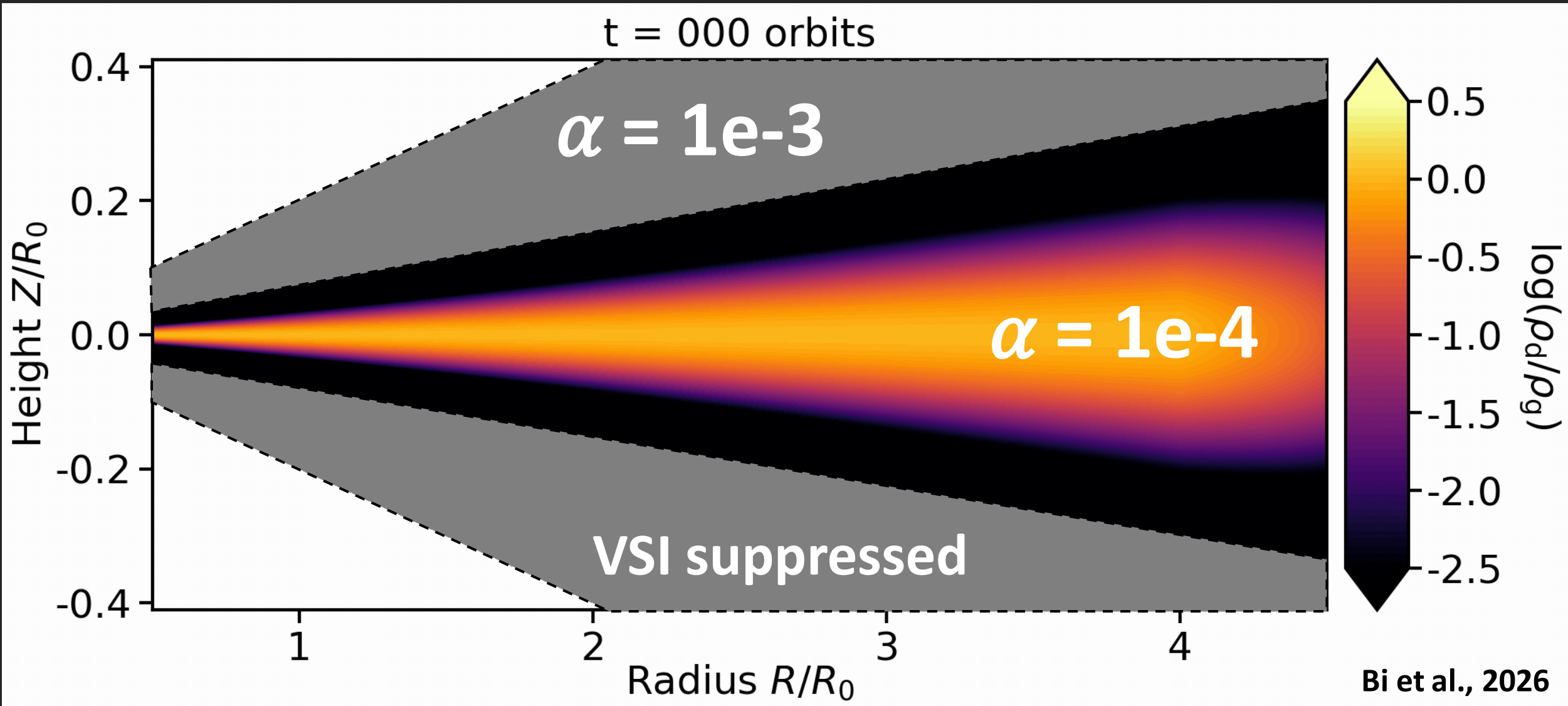




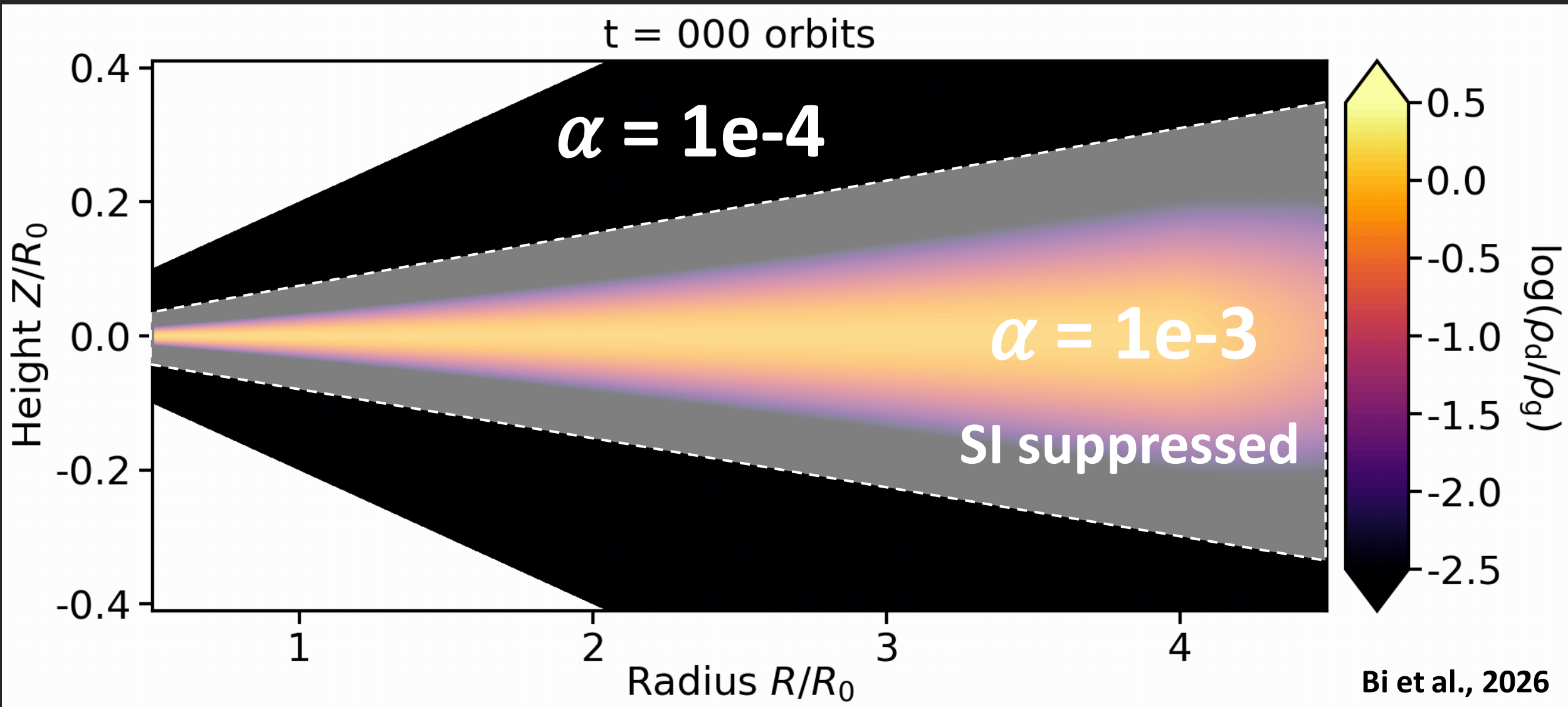
Mask atmosphere w/ high viscosity ($\alpha = 1e-3$)



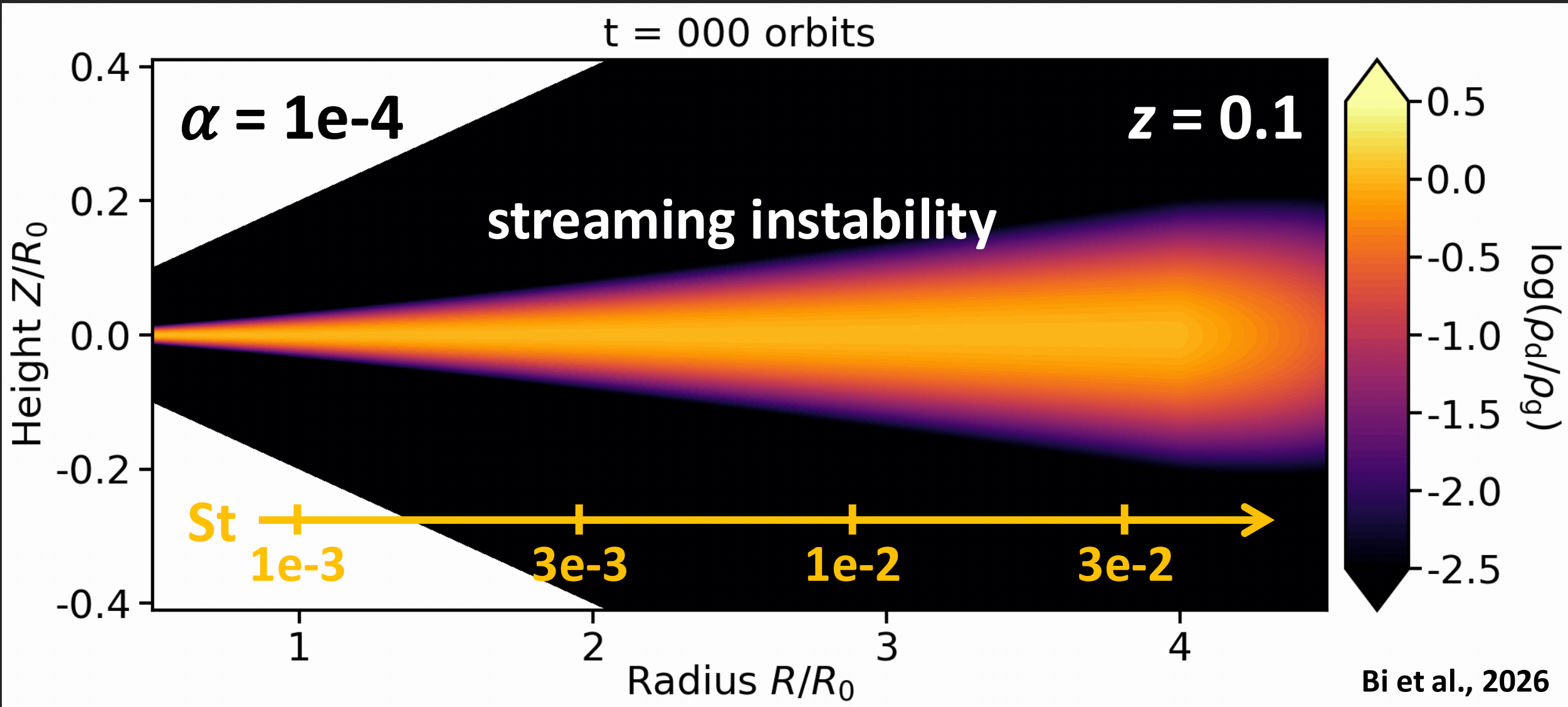
Mask atmosphere w/ high viscosity ($\alpha = 1e-3$)



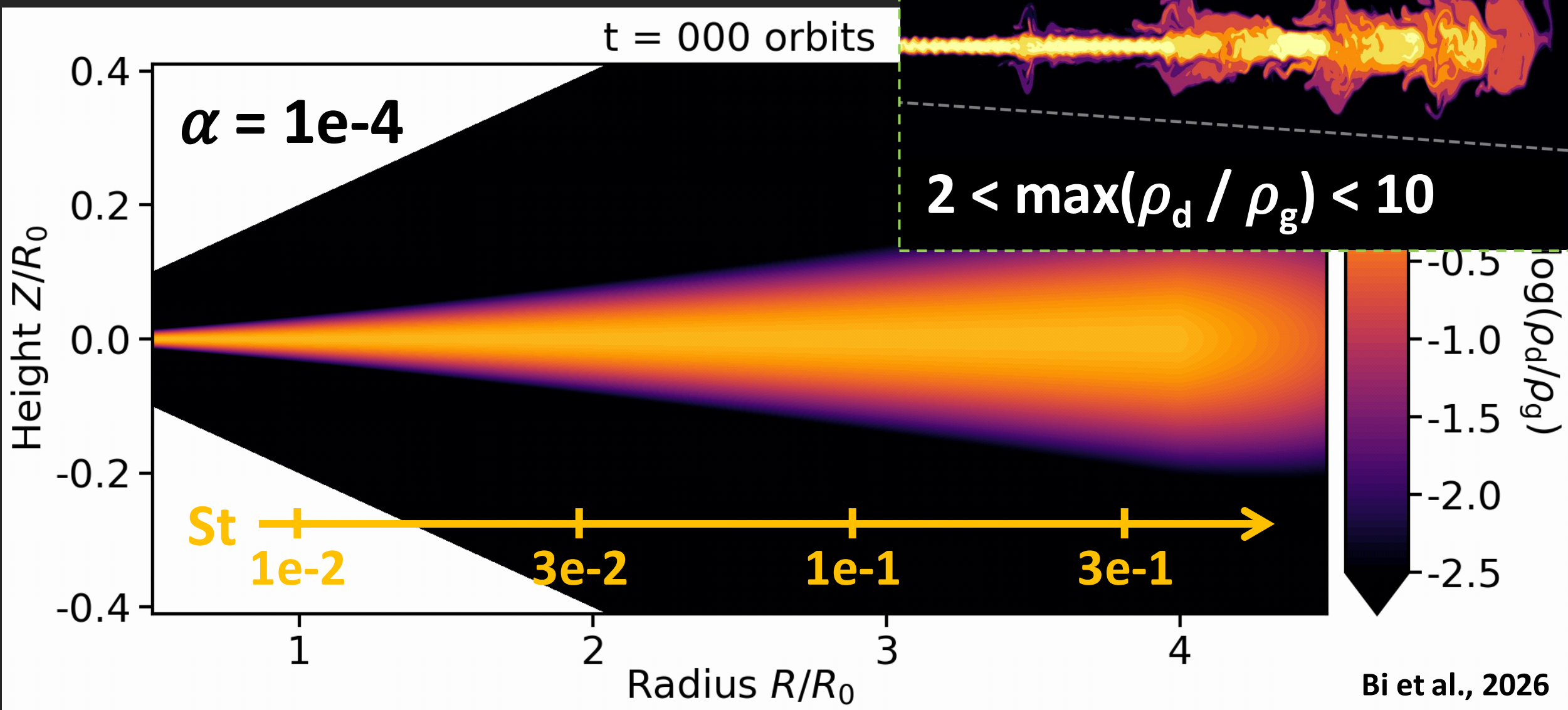
Mask midplane w/ high viscosity ($\alpha = 1e-3$)



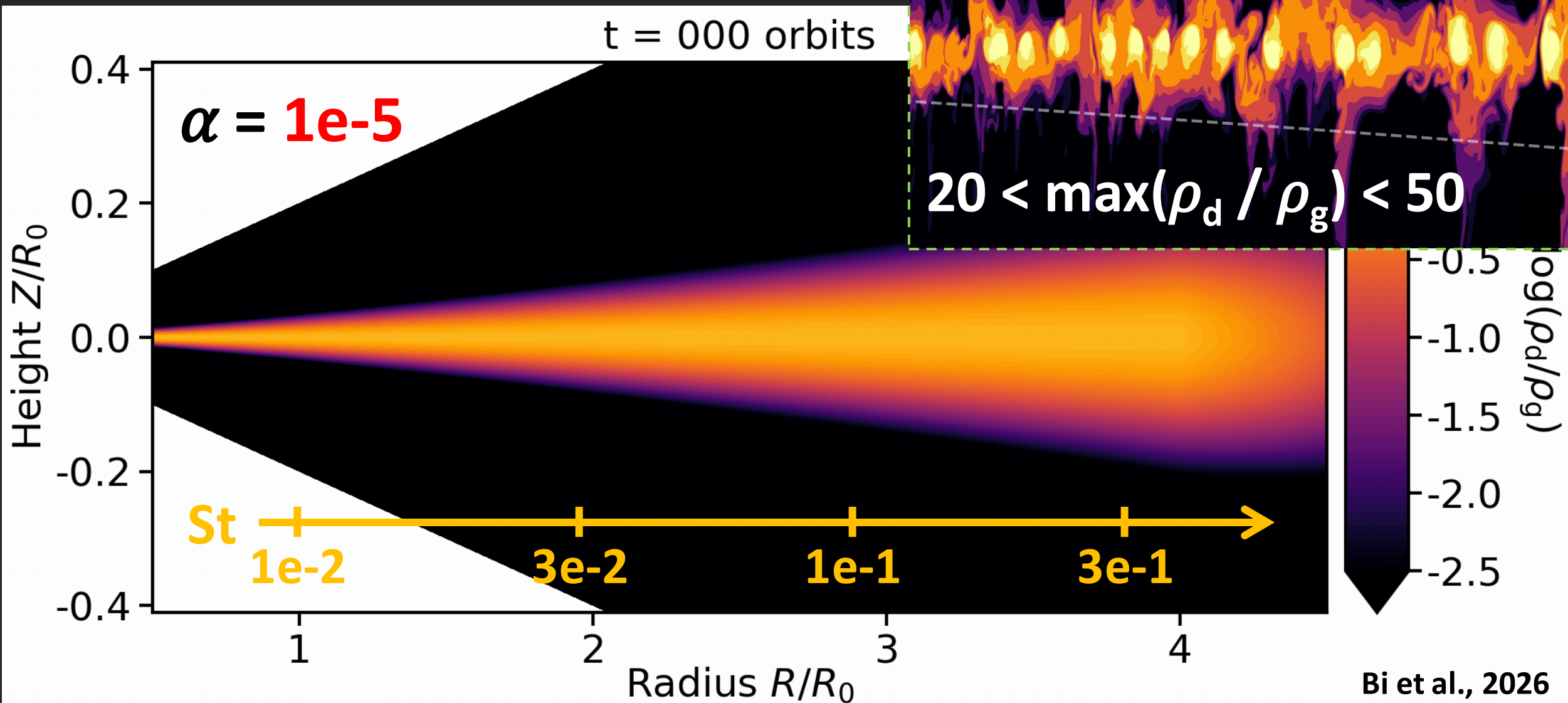
Substructures Induced by Dust Drag



Larger grains in viscous gas



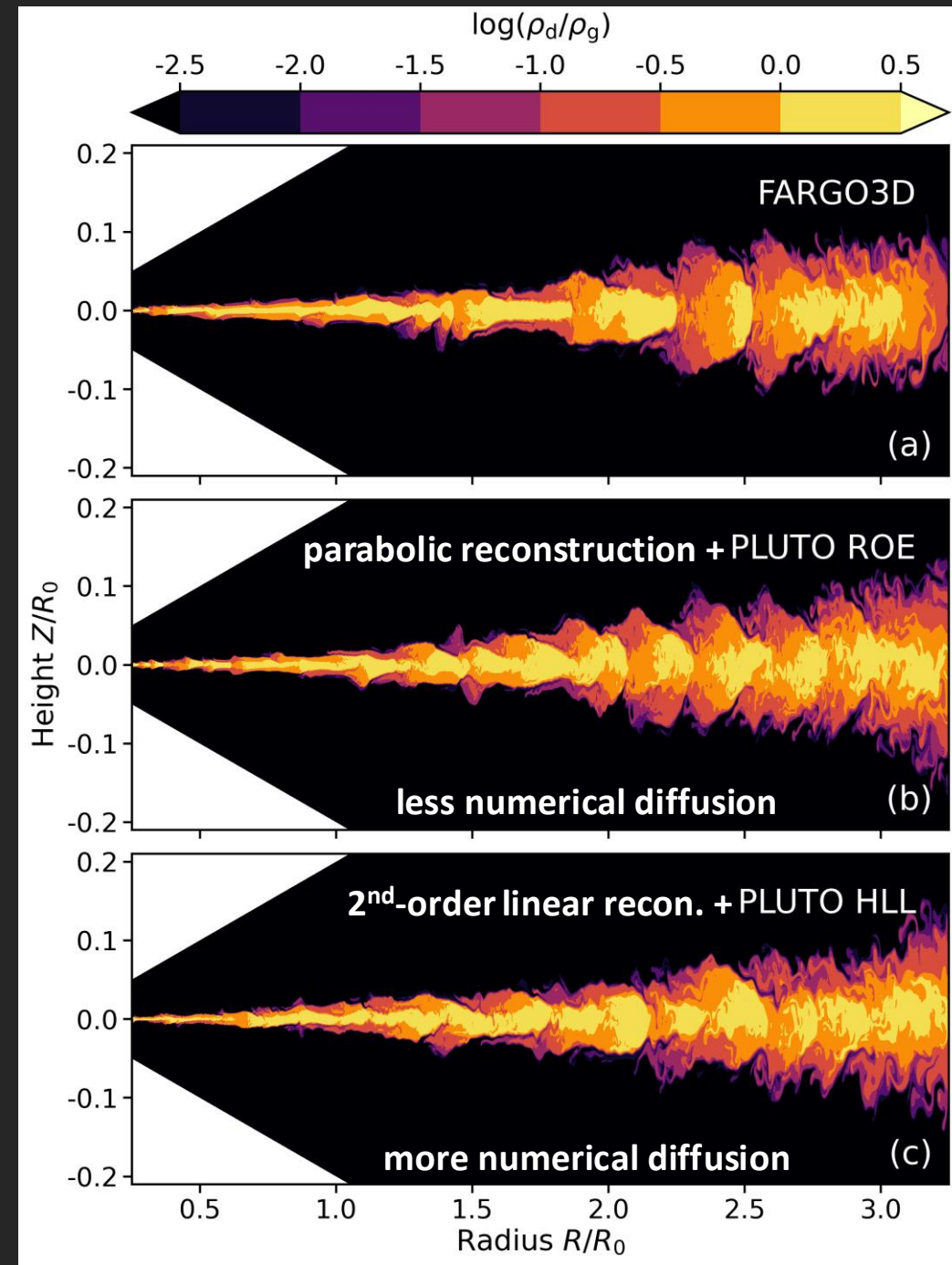
Larger grains w/ low viscosity

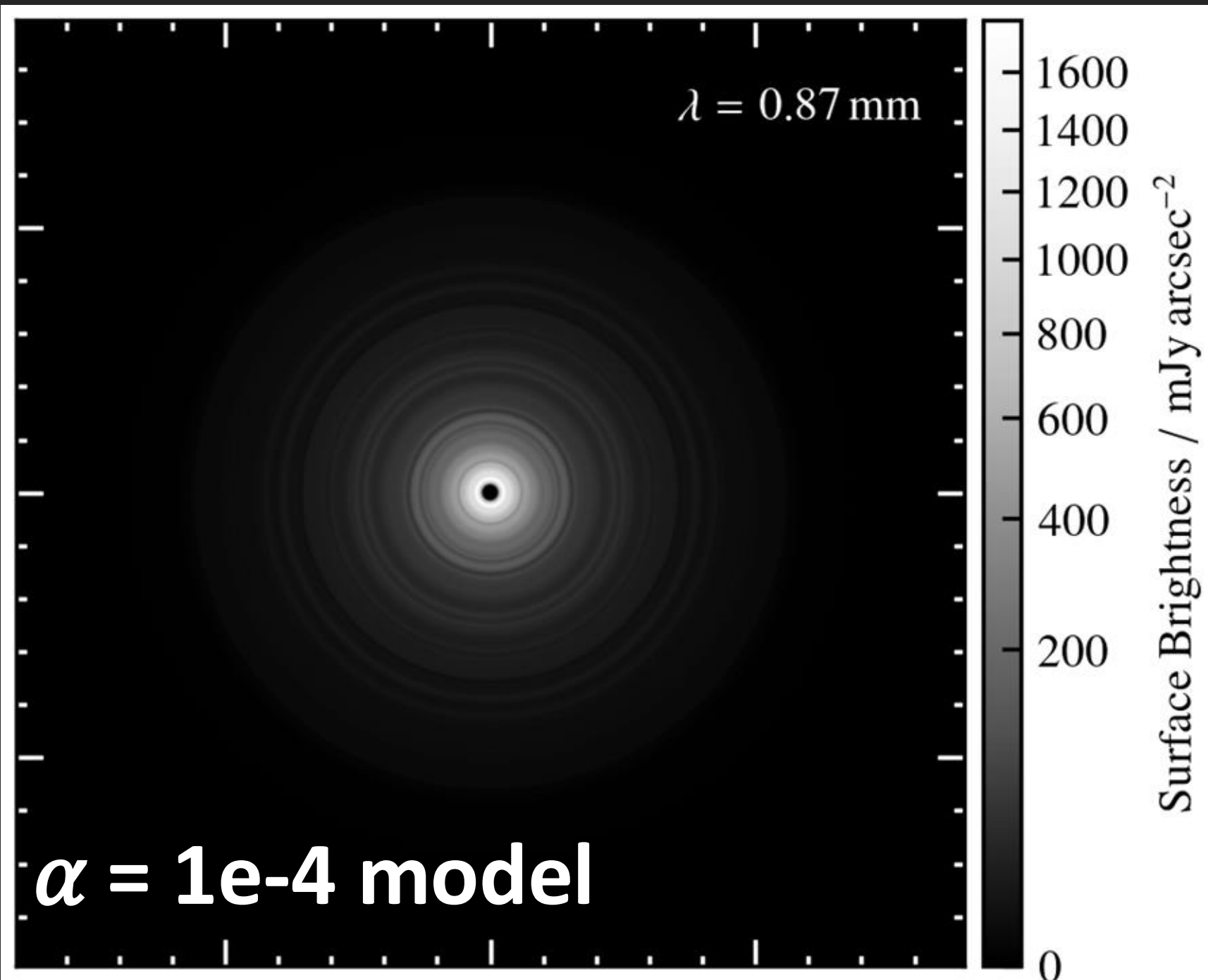


Numerical effect? Maybe not

The “birdie-like” substructure

- is primarily driven by SI
- only shows up within a certain viscosity range around $\alpha = 1e-4$
- is robust across different integration and reconstruction methods





Observability?

**w/ S. Wolf's team
at Kiel University**

**mechanical
alignment
vs.
self-scattering**

stay tuned